Chapter 7
Swaps and the Multi-Curve Framework

The interest rate swap is the largest derivatives market in the world by notional outstanding, and it is the instrument through which almost every other rate derivative is hedged. It is also, for our purposes, the place where a simplification made in Chapter 4 has to be paid for.

That chapter spoke of "the" zero curve for a currency, as though a single term structure described it. The assumption was flagged as provisional, and this chapter retires it. A modern swap valuation requires *at least two* curves: one that says what cash flows are worth, and one that says how large the floating cash flows will be. They are built from different instruments, they move independently, and the gap between them is traded in large volume.

It would be easy to present this as a technical correction added to a simpler theory—which is how it entered the market, in the years after 2008, and how most textbooks still frame it. That framing is misleading. The single-curve world was not a simpler truth later refined; it was a coincidence. Two things happened to line up: the rate at which banks lent to each other for three months, and the rate at which a derivative's collateral earned interest. Once those separated, they never rejoined, and there is no reason they should have. This chapter therefore develops the two-curve structure as the standard case and treats the single-curve world as the special case it always was.

Sections 7.1 to 7.3 are the contract, why it exists, and how it is valued. Sections 7.4 to 7.6 are the two-curve structure: why the collateral agreement determines the discount curve, why one curve is not enough, and how the curves are built. Sections 7.7 to 7.9 are the instruments and prices that structure produces (basis swaps, cross-currency swaps and their basis, and swap spreads as the price of balance sheet), and Sections 7.10 to 7.12 close with credit, netting and clearing, what the transition did to the swap book, and the other swaps the market trades.

7.1 MECHANICS OF AN INTEREST RATE SWAP

A **swap** is an agreement to exchange cash flows on a schedule. In the standard interest rate swap, the **plain vanilla** swap, two parties agree to exchange, on a notional principal L, a stream of fixed interest payments for a stream of floating payments. The notional is never exchanged; it exists only to scale the payments, which is why an enormous notional outstanding corresponds to a much smaller amount of money at risk.

Since the transition described in Chapter 4, the floating leg of a standard swap in dollars, sterling, Swiss francs, yen and Canadian dollars references a compounded risk-free rate in arrears. Such a contract is the overnight indexed swap of Section 4.8: a **SOFR OIS** in dollars, a SONIA OIS in sterling, and so on for each currency's designated rate. In those currencies it is now the market standard rather than a specialist product. The euro is the large exception: EURIBOR survived the transition (Section 4.3), and most euro swaps still pay an annual fixed rate against six-month EURIBOR, with €STR OIS trading beside them; Australian, Hong Kong and Chinese swaps likewise still reference term rates. In dollars the name OIS once

meant a different contract. Before October 2020 the OIS curve was built from federal funds swaps, not SOFR, and Section 7.4 describes the switch; where this chapter writes OIS without qualification, the currency's current rate is meant. On each payment date the floating payer owes

$$L\,\tau_i\,R_i \tag{7.1}$$

where τ_i is the length of accrual period i as a fraction of a year, R_i is the compounded RFR over that period computed as in Section 4.4, and the fixed payer owes $L\,\tau_i\,K$ for the agreed fixed rate K. Only the difference changes hands. Figure 7.1 shows the arrangement.



Figure 7.1: Cash flows in a plain vanilla interest rate swap. The notional L is not exchanged.

Example 7.1 A swap's cash flows over its life

Company B enters a three-year annual swap on a notional of \$100 million, paying 4.20% fixed and receiving the compounded RFR. Suppose the compounded rate turns out to be 4.00% over the first year, 4.35% over the second, and 4.60% over the third. All figures below are in millions of dollars.

Payment date	Compounded RFR	Floating received	Fixed paid	Net to B
End of year 1	4.00%	+4.00	−4.20	−0.20
End of year 2	4.35%	+4.35	−4.20	+0.15
End of year 3	4.60%	+4.60	−4.20	+0.40
Total				**+0.35**

Three features of the table repay attention.

Only the net column changes hands: the gross \$4 million legs are never exchanged, and neither is the \$100 million notional.

The rate in the second column is not known at the start of the period it applies to. Under the conventions of Chapter 4 it accumulates from daily fixings across the year and is complete only at the end of the period, a few days at most before it is paid. Under the LIBOR conventions this table would have replaced (where each rate was fixed twelve months in advance), B would have known each payment a year ahead. This is the single largest operational change the transition brought.

B did well here only because rates rose. Had they fallen to 3.5%, B would have paid \$700,000 in that year. That is not a loss in any meaningful sense: B entered the swap to *fix* its cost, and Example 7.2 shows the combined position doing exactly that.

Conventions. A swap confirmation specifies far more than the fixed rate: the payment frequency on each leg (usually annual on both legs for an OIS, while swaps on the surviving term rates mix frequencies: annual fixed against semiannual EURIBOR in euros, quarterly against quarterly BBSW in Australian dollars), the day count on each leg, the business day convention, the calendar, and, critically after Chapter 4, the exact compounding convention and payment lag on the floating leg. Two swaps quoted at the same rate under different conventions are not the same trade, and the difference is not always small. Table 7.1 sets out a confirmation for a five-year dollar swap, and later sections return to it: Section 7.3 values it, and Section 7.4 shows that its last line, the collateral, is what fixes the curve used to do so.

The fixed rate is 4.31% rather than the 4.295% midpoint of the dealer's quote because the customer is paying fixed and therefore deals at the higher of the dealer's two rates, as the next

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Table 7.1: Terms of a five-year dollar swap. Illustrative, not a market quote.

Notional	$100,000,000
Effective date	March 18, 2026
Maturity date	March 18, 2031
Fixed rate	4.31%, paid by the customer
Fixed leg	Annual, actual/360
Floating index	SOFR, compounded daily in arrears
Floating leg	Annual, actual/360
Payment lag	Two business days
Business day convention	Modified following
Calendar	New York
Clearing	Cleared at a CCP
Collateral	Variation margin in USD cash, interest at SOFR

subsection explains. Every other line is a convention, and every one of them changes the cash flows.

How swaps are quoted

A dealer quotes a swap by quoting the *fixed* rate; the floating leg is understood to be the index flat, with no spread. A quote of "4.28 / 4.31" on the five-year means the dealer will receive fixed at 4.31% and pay fixed at 4.28%, and the three-basis-point difference is the bid–ask spread. The customer pays a slightly higher fixed rate than the mid when paying fixed and receives a slightly lower one when receiving. This is the usual arrangement, with the notable feature that the spread is quoted in rate rather than in price. Three basis points sounds negligible and is not: on the notional of Table 7.1 it is worth about $134,000 over the life of the trade, by the calculation Section 7.3 sets out.

Swaps are often traded on the IMM dates of Section 6.2, the standardized quarterly dates used by futures markets, so that a swap position can be hedged against futures without a date mismatch.

A particular class of standardized contracts, **market agreed coupon** swaps, fixes the coupon at a round number in increments of 25 basis points and settles the difference from the par rate as an upfront payment. These are one product among many rather than the general case: the swap of Table 7.1 is an ordinary par swap, struck at whatever rate makes it worth zero, and most of the market trades that way. What the round coupon buys is *fungibility*. Two par swaps struck a day apart carry different fixed rates and net against each other only in value; two market agreed coupon swaps of the same coupon and dates are the same instrument, so they compress and clear more cheaply. That matters most to dealers and to hedge funds and other active traders, who open and close positions often and want them to net; a company hedging one particular loan has no use for fungibility and trades a par swap matched to the loan. What it costs is cash on day one. Written that way, Table 7.1's swap would carry a 4.25% coupon (the nearest quarter-point to its 4.295% par rate), and the 4.5 basis point difference, multiplied by the swap's annuity of 4.48 (the present value of the year fraction accrued in each of its five years, about 1.014 under actual/360, discounted at the par rate; Section 7.3), gives

$$\$100{,}000{,}000 \times 0.00045 \times 4.48 = \$201{,}600$$

half as much again as the entire bid–ask. Because the coupon is the nearest quarter-point and not the one below, the payment runs in either direction: here the fixed payer makes it, having been given a coupon below the market rate, but at a par rate of 4.205% the same coupon would sit above par and the fixed payer would receive it. Half an increment also bounds it at inception (about $560,000 on this trade), though a standing coupon the market has since moved away

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from can exceed that.

The dealer running such a book is not taking a view. Having received fixed from one customer, it hedges by paying fixed elsewhere: to another customer, in the interdealer market, or with futures. What remains after hedging is a residual exposure to the shape of the curve, to the basis spreads of Section 7.5, and to the counterparty credit of Section 7.10. Managing those residuals, rather than predicting rates, is the business. The residual is measured bucket by bucket, as the dollar value of one basis point (DV01) of each quoted maturity separately: the book's change in value for a one-basis-point move in that maturity alone. This bucketed or key-rate DV01 shows exposure to the shape of the curve that a single parallel-shift number nets away. The short buckets are typically hedged with the futures of Chapter 6, the longer ones with swaps.

7.2 WHY SWAPS EXIST

A swap changes the interest rate character of an exposure without disturbing the contract that created it. That single sentence covers most of the market, but it is worth working through the four standard uses concretely, because the cash flow accounting is what makes the idea concrete.

Transforming a liability

This is the most common use. A company has borrowed at a floating rate and wants certainty about its interest cost. Figure 7.2 shows the three flows and where the floating rate cancels.

Example 7.2 Converting floating debt to fixed

Company B has a three-year loan of $50 million on which it pays the RFR plus 80 basis points. It would rather know its cost. It enters a swap on the same notional, paying 4.20% fixed and receiving the RFR.

Its three cash flows are now:

Pays on the loan	−(RFR + 0.80%)
Pays on the swap	−4.20%
Receives on the swap	+RFR
Net cost	−5.00% fixed

The RFR appears twice with opposite signs and cancels. B has converted a floating liability into a fixed one at 5.00%, or $2.5 million a year, without renegotiating the loan or informing the lender.



Figure 7.2: Company B pays its lender on the left and swaps on the right.

Note what has *not* changed. B still owes the loan, still bears the 80 basis point credit spread, and, as Section 7.10 discusses, has added a counterparty to its balance sheet. What it has removed is one specific risk: the level of the reference rate.

Transforming an asset

The same operation applies to the other side of the balance sheet, and its diagram is the mirror image of Figure 7.2, with every arrow reversed. An investor holding a floating-rate asset who wants a fixed return receives fixed and pays floating.

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Example 7.3 Converting a floating-rate asset to fixed

Company A holds \$50 million of floating-rate notes yielding the RFR plus 25 basis points, and wants a fixed return to match a fixed obligation. It enters a swap receiving 4.20% fixed and paying the RFR:

Receives on the notes	+(RFR + 0.25%)
Receives on the swap	+4.20%
Pays on the swap	−RFR
Net return	+4.45% fixed

A has synthesized a 4.45% fixed-rate asset out of a floating one.

These two examples are mirror images, and in principle A and B could face each other directly. In practice they do not, for the obvious reason that they do not know each other and would not want each other's credit.

The role of the financial intermediary

A dealer stands between them, as in Figure 7.3 and Example 7.4, entering offsetting swaps with each and quoting a bid–ask spread on the fixed rate, say 4.18 / 4.22, read as in Section 7.1: the dealer pays fixed at 4.18% and receives fixed at 4.22%.

Example 7.4 The dealer in the middle

The dealer pays A 4.18% fixed and receives the RFR; it receives 4.22% fixed from B and pays the RFR. The floating legs offset exactly, so the dealer is left with

$$4.22\% - 4.18\% = 4 \text{ basis points}$$

on \$50 million, or \$20,000 a year, for as long as both swaps run.

The all-in outcomes shift slightly. A now earns 4.18% + 0.25% = 4.43% rather than 4.45%, and B pays 4.22% + 0.80% = 5.02% rather than 5.00%. Between them they have paid the dealer four basis points for finding the other side and standing between them.

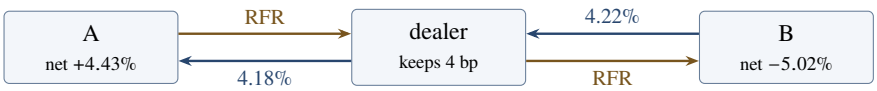


Figure 7.3: The dealer faces both customers. Its two floating legs offset, leaving the four basis point spread.

The dealer is not merely a matchmaker. It takes B's credit risk and A's credit risk, and it need not have found A and B simultaneously: in practice it warehouses one side, hedges the rate risk in the futures market of Chapter 6, and waits for offsetting business. What it earns is compensation for that warehousing and for the credit it has taken, not a broker's fee.

The comparative advantage argument, and what is wrong with it

The fourth traditional motivation is comparative advantage, and it deserves both a fair statement and a critique.

Suppose Company A and Company B can each borrow for five years in either market, on these terms:

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	Fixed	Floating
Company A (stronger credit)	5.0%	RFR + 0.3%
Company B (weaker credit)	6.4%	RFR + 1.0%
Difference	1.4%	0.7%

A is cheaper in both markets, but its advantage is larger in the fixed market (1.4%) than in the floating market (0.7%). Suppose further that A *wants* floating and B wants fixed, which is the inconvenient case, since each wants the market in which it is relatively worse off.

The apparent total gain is $1.4\% - 0.7\% = 0.7\%$. Splitting it evenly, each should end up 35 basis points better off than by borrowing directly. Example 7.5 shows how.

Example 7.5 Splitting the comparative advantage gain

A borrows fixed at 5.0%, its stronger market; B borrows floating at RFR + 1.0%. They then swap, with A receiving 5.0% fixed and paying RFR − 0.05%.

Company A:

$$\underbrace{-5.0\%}_{\text{loan}} + \underbrace{5.0\%}_{\text{swap received}} - \underbrace{(\text{RFR} - 0.05\%)}_{\text{swap paid}} = -(\text{RFR} - 0.05\%)$$

A borrows floating at RFR − 0.05%, against RFR + 0.30% directly, better by 35 basis points.

Company B:

$$\underbrace{-(\text{RFR} + 1.0\%)}_{\text{loan}} - \underbrace{5.0\%}_{\text{swap paid}} + \underbrace{(\text{RFR} - 0.05\%)}_{\text{swap received}} = -6.05\%$$

B borrows fixed at 6.05%, against 6.4% directly, also better by 35 basis points.

The two gains sum to the 0.7% predicted, and the RFR cancels for both parties.

The arithmetic is correct. The economics is not, and it is worth seeing why, because the flaw illustrates something general about apparent riskless profits.

The two rate pairs are not comparable commitments. A five-year fixed loan is a five-year commitment by the lender. A floating-rate loan is typically rolled at intervals, and the spread over the index is reset (or the facility withdrawn) if the borrower's credit deteriorates. The floating lender therefore has an option the fixed lender does not, and the weak borrower B pays a smaller *quoted* spread in the floating market precisely because the lender retains the right to walk away.

When B enters a swap to convert its floating borrowing into synthetic fixed, it does not thereby acquire a five-year fixed rate. It acquires a five-year swap plus a rolling floating loan that may be repriced against it. The "gain" is compensation for retaining exactly the risk that made the floating quote look cheap. If B's credit deteriorates, the swap keeps paying but the loan spread widens, and the synthetic fixed rate rises.

Comparative advantage is therefore a poor explanation of why the swap market is the size it is. The better explanation is the mundane one contained in Examples 7.2 and 7.3: swaps let institutions separate the interest rate character of an exposure from the funding decision that created it, and they do so with far less balance sheet than transacting in the underlying bonds would require.

Other users

Beyond the four classical uses, four groups account for much of the market's size.

Banks managing the gap. A bank whose deposits reprice overnight while its loan book or securities portfolio is fixed is structurally exposed to rates rising, and closes that mismatch

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with swaps rather than by changing its lending. This is one of the largest sources of end-user demand. Silicon Valley Bank in March 2023 is the case of a bank that left the mismatch largely open: it held long fixed-rate securities against deposits that could leave overnight, the 2022 rise in rates cut the securities' value well below cost, and the deposits left when depositors saw it.

Mortgage hedgers. A mortgage servicer or a holder of mortgage-backed securities loses duration when rates fall, because borrowers refinance, and gains it when rates rise. To stay hedged it must pay fixed in swaps as rates rise and receive as they fall, which is buying high and selling low on a large scale; in a sharp move that flow pushes swap rates further in the direction they were already going.

Pension funds and insurers matching liabilities. Long-dated liabilities discounted at market rates create enormous duration exposure. Receiving fixed in long-dated swaps lengthens asset duration toward liability duration without buying bonds, which is why, as Section 7.9 notes, this demand compresses swap spreads, and why the 2022 crisis in UK pension funds' liability-driven investment (LDI) strategies, which Chapter 8 examines, was driven by collateral calls on swap and repo positions rather than by losses on bonds held outright.

Issuers arbitraging their own funding. A company that can issue a fixed-rate bond cheaply but prefers floating exposure issues the bond and swaps it. This is what remains of the comparative advantage idea: the gain is real, but it comes from the issuer's access to a particular investor base, not from a mispricing between two markets.

What makes all of this practical for a nonfinancial company is hedge accounting. A swap is a derivative and would otherwise be marked to market through earnings each quarter while the debt it hedges sits at cost, so the hedged position would show volatility it does not have. Under IFRS 9 and its US counterpart, ASC 815 (Section 3.8), a swap documented as a hedge of specific debt has its gains and losses recognized alongside the hedged item, and reported earnings show the fixed or floating cost the company chose.

7.3 VALUING A SWAP

There are two equivalent ways to value a swap, and it is worth having both.

As an exchange of bonds. A swap is a long position in one bond and a short position in another. The fixed payer is short a bond paying coupon K and long a floating-rate note.

As a portfolio of forward contracts. Each exchange of payments is an FRA of the kind valued in Section 4.9. The swap is their sum. This is the more useful view, because it extends directly to the multi-curve case.

Take the second route, and write DF_i for the discount factor to payment date t_i and F_i for the forward rate for accrual period i. The ith exchange alone is an FRA: the fixed payer owes $L\tau_i K$ and is owed $L\tau_i R_i$, and valuing it means substituting the forward rate F_i for the unknown R_i and discounting, so it is worth

$$f_i = L\,\tau_i\,(F_i - K)\,\mathrm{DF}_i \tag{7.2}$$

which is the FRA value of Section 4.9 seen from the other side, there written for the party receiving fixed, here for the party paying it. Summing (7.2) over the payment dates gives the swap:

$$V_{\text{payer}} = L\sum_{i=1}^{n}\tau_i\,F_i\,\mathrm{DF}_i \;-\; L\,K\sum_{i=1}^{n}\tau_i\,\mathrm{DF}_i \tag{7.3}$$

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The second sum is the **annuity**

$$A = \sum_{i=1}^{n} \tau_i \, \mathrm{DF}_i \tag{7.4}$$

the present value of one unit of fixed rate received on the swap's schedule; multiplied by $0.0001L$ it is the present value of one basis point of fixed rate on the notional, which is why it is also called the PV01 leg. It will be used repeatedly: it is what converts a difference between rates into an amount of money, it is the swap's sensitivity to its own fixed rate, and Chapter 29 will take it as a numeraire (the unit in which other prices are measured) in its own right. A five-year annual swap discounted at 4%, with each year fraction taken as one, has $A \approx 4.45$, so a one-basis-point move in the par rate changes the value of \$100 million of notional by about \$44,500.

Setting $V = 0$ and solving gives the **par swap rate**, the fixed rate at which a new swap has zero value:

$$S = \frac{\sum_{i=1}^{n} \tau_i \, F_i \, \mathrm{DF}_i}{\sum_{i=1}^{n} \tau_i \, \mathrm{DF}_i} \tag{7.5}$$

so the par rate is a weighted average of the forward rates, with the discount factors as weights. From (7.3) and (7.5) the value of an existing swap has the compact form

$$V_{\text{payer}} = L \, (S - K) \, A \tag{7.6}$$

which says that a swap is worth the difference between the current par rate and its own fixed rate, annuitized. Most swap risk management reduces to this expression.

A telescoping identity—and a warning

Something convenient happens if the floating index and the discount curve come from the *same* curve. In that case the forward rate satisfies $\tau_i F_i = \mathrm{DF}_{i-1}/\mathrm{DF}_i - 1$ by definition, with $\mathrm{DF}_0 = 1$ for today, so

$$\sum_{i=1}^{n} \tau_i F_i \, \mathrm{DF}_i = \sum_{i=1}^{n} (\mathrm{DF}_{i-1} - \mathrm{DF}_i) = 1 - \mathrm{DF}_n \tag{7.7}$$

The whole floating leg collapses to $L(1 - \mathrm{DF}_n)$: the forward rates cancel, and only the first and last discount factors survive. This is the classic result that a floating-rate note prices at par on its reset dates.

Remember this identity, and remember equally that it is *false in general*. It holds only when the rate being forecast is the same rate being used to discount. That was approximately true before 2008 and is not true now. Section 7.5 shows what replaces it.

Example 7.6 Valuing an existing swap

A three-year annual swap on a notional of \$100 million pays fixed at 4.00% and receives the compounded RFR. Using the OIS curve bootstrapped in Chapter 4, the discount factors are 0.961538, 0.920936, and 0.881164, and the corresponding forward rates are 4.0000%, 4.4088%, and 4.5136%.

Floating leg. Discounting each projected payment,

$$100\big[0.040000 \times 0.961538 + 0.044088 \times 0.920936 + 0.045136 \times 0.881164\big] = \$11{,}883{,}596$$

The telescoping identity (7.7) gives the same answer in one step, to the rounding in the forward rates, since index and discount curve agree here:

$$100\,(1 - 0.881164) = \$11{,}883{,}600$$

Fixed leg. The annuity is $A = 0.961538 + 0.920936 + 0.881164 = 2.763638$, so the fixed leg

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is worth

$$100 \times 0.04 \times 2.763638 = \$11{,}054{,}552$$

Value. To the fixed payer the swap is worth 11,883,600 − 11,054,552 = \$829,048.

Checking with (7.6): the par rate is $(1 - 0.881164)/2.763638 = 4.30\%$, so the swap is worth $100 \times (0.0430 - 0.0400) \times 2.763638 = \$829{,}091$, agreeing to rounding. The payer is receiving a floating stream now worth 4.30% while paying only 4.00%.

7.4 THE COLLATERAL AGREEMENT DETERMINES THE DISCOUNT CURVE

We now come to the question the previous section left open: discount at *what*?

The answer is not a matter of preference. Consider a swap that is fully collateralized under a CSA, the standard arrangement described in Chapter 8. Each day, the party for whom the swap has negative value posts collateral equal to that value, and receives interest on the posted collateral at a rate specified in the CSA, almost always the overnight rate of the collateral currency.

Now think about what a dealer holding such a position actually owns. If the swap's value rises by \$1, the dealer receives \$1 of collateral, on which it owes the poster the overnight rate until the position reverses. The dealer may use that cash, so the gain on the swap is financed by a liability that accrues at the collateral rate: the position is, in cash flow terms, self-financing at the overnight rate. Replicating it therefore costs the overnight rate, and by the arbitrage arguments of Chapter 5, the correct discount rate for a fully collateralized derivative is **the rate paid on its collateral**.

This has a consequence that surprises people the first time they meet it: the same contract, with the same cash flows, has a different value under different CSAs. A swap collateralized in dollars earning SOFR and an otherwise identical swap collateralized in euros earning €STR are not worth the same. The CSA is part of the instrument.

A cleared swap, such as the one in Table 7.1, works the same way with the clearing house's rulebook in place of a CSA: variation margin is paid in cash and earns price alignment interest at the overnight rate, so cleared dollar swaps are discounted on the SOFR curve. Bilateral CSAs change more slowly, and some older dollar CSAs still pay the effective federal funds rate; trades under them are discounted on a federal funds curve, a live case of the same contract carrying two values.

Case Study 7.1 The day discounting changed

For years after the crisis, cleared swap markets discounted at the effective federal funds rate in dollars and the euro overnight index average (EONIA) in euros, because those were the rates paid on cleared collateral. When the market moved to SOFR and €STR, the collateral rate changed—and therefore so did the value of every cleared swap.

The clearing houses did not simply re-mark and move on. LCH (originally the London Clearing House) and CME executed the dollar switch over a weekend in October 2020, changing the discounting rate on their entire cleared portfolios at once. Because the change transferred value between members (some portfolios gained, others lost), each house paid cash compensation to make members whole, and additionally distributed basis swaps to offset the change in risk profile that the discounting switch produced. The euro switch, in July 2020, needed only the cash: EONIA had been redefined in 2019 as €STR plus a fixed 8.5 basis points, so the spread between the old and new discount rates was known for every future date and left no basis risk to hand back.

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> Two lessons follow. First, the choice of discount curve is not an accounting convention; it is worth enough money that changing it required a coordinated, compensated, market-wide operation. Second, discounting risk is a real risk: a portfolio can be flat to rates and still gain or lose from a change in the rate its collateral earns.

Uncollateralized trades. If a swap is not collateralized, as many corporate trades are not, the replication argument above does not apply, because the dealer must fund the position from its own resources. The discount rate then involves the dealer's funding cost, which introduces the funding valuation adjustment of Chapter 9. It also means the trade's value depends on *who is holding it*, which is uncomfortable but true.

7.5 WHY ONE CURVE IS NOT ENOUGH

We now have two distinct roles for a curve, and no reason for one curve to fill both.

- The **discount curve** answers: what is a cash flow at time t worth today? Section 7.4 settled this: it is determined by the collateral rate in the CSA, and is built from OIS.
- A **forecast curve** answers: how large will the floating payment be? This is determined by whatever index the contract references, which may be the same overnight rate, or term SOFR, or a foreign currency's RFR, or a legacy fallback rate.

Figure 7.4 sets out the division of labor. When the index and the collateral rate coincide, the two curves coincide and everything collapses to the single-curve formulas, including the telescoping identity (7.7). When they do not, the curves must be built and used separately, and Example 7.7 measures the error of letting one curve do both jobs.

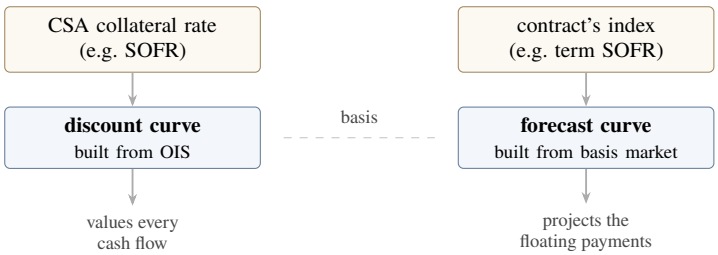


Figure 7.4: The two roles a curve plays. Before 2008 the same curve filled both.

The basis is not noise

The gap between the two curves is the **curve basis**, and it is a traded quantity with its own market. Historically the largest example was the spread between LIBOR and OIS, which measured the credit and liquidity premium in interbank lending. Before 2007 it sat at a small number of basis points and was widely ignored; in late 2008 it exceeded 350 basis points. A dealer valuing a swap book on the assumption that the two were the same would have been wrong by an amount that dwarfed every other modeling choice in the portfolio.

That particular basis has gone with LIBOR. Others remain and are structural rather than crisis-driven:

- **Term versus compounded.** Term SOFR is a forward-looking rate derived from futures; compounded SOFR is backward-looking. They are not equal, and the difference is traded,

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though narrowly: under the ARRC's scope-of-use rules, term SOFR derivatives are confined to end users hedging loans and other cash products that reference it, and the dealers who face them lay the exposure off in term SOFR against SOFR basis swaps.

- **Tenor basis.** Where a currency retains term indices of different lengths, a swap referencing the shorter index against one referencing the longer trades at a spread. The euro is the main case: three-month against six-month EURIBOR is quoted as a basis swap (the "3s6s" basis), and euro swap systems carry a separate forecast curve for each tenor. The economics is the same as the old LIBOR tenor basis: rolling a short index carries different risk than fixing a long one.
- **Cross-currency basis.** The subject of Section 7.8.

Example 7.7 Where the single-curve identity breaks

Take the swap of Example 7.6, but suppose the floating leg references an index whose forward rates sit 10 basis points above the OIS forwards (a term rate, say). The forecast forwards become 4.1000%, 4.5088%, and 4.6136%, while the discount factors, being set by the CSA, are unchanged.

The floating leg is now

$$100\big[0.041000 \times 0.961538 + 0.045088 \times 0.920936 + 0.046136 \times 0.881164\big] = \$12{,}159{,}960$$

against \$11,883,596 from the same calculation in Example 7.6. The difference of \$276,364 is exactly the basis times the annuity:

$$100 \times 0.0010 \times 2.763638 = \$276{,}364$$

The telescoping identity (7.7) would have given \$11,883,600 and been wrong by more than a quarter of a million dollars—because the rate being forecast is no longer the rate being used to discount. The par swap rate rises from 4.30% to 4.40%: the 10 basis point basis passes through one-for-one into the par rate.

7.6 BUILDING THE CURVES

Curve construction in a multi-curve world proceeds in a definite order, because the pieces depend on one another.

Step 1: the discount curve. Bootstrap from OIS quotes, exactly as in Section 4.8. These instruments have a single index, and that index is the collateral rate, so the construction is self-contained.

Step 2: the forecast curves. For each index the book references, build a curve that reprices the instruments referencing that index, while discounting them on the curve from Step 1. This is the step that makes the procedure genuinely multi-curve: the forecast curve is defined by what it must reprice, given a discounting choice already fixed.

Step 3: check consistency. Basis swaps quoted between two indices must be repriced by the pair of forecast curves. In practice this is not a sequential bootstrap but a simultaneous calibration: the curves are solved jointly so that every quoted instrument (OIS, index swaps, basis swaps, futures with their convexity corrections from Chapter 30) is repriced at once. Example 7.8 carries the three steps through by hand.

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Example 7.8 Two curves from two sets of quotes

Payments are annual and $\tau_i = 1$ throughout, so the construction can be followed by hand.

Step 1: the discount curve. OIS quotes are 4.00%, 4.20%, and 4.30% for one, two, and three years. An OIS forecasts the same rate it discounts at, so its floating leg telescopes and the par condition is simply $S \sum_i \mathrm{DF}_i = 1 - \mathrm{DF}_n$. Solving in order,

$$\mathrm{DF}_1 = \frac{1}{1.04} = 0.961538, \qquad \mathrm{DF}_2 = \frac{1 - 0.042 \times 0.961538}{1.042} = 0.920936$$

$$\mathrm{DF}_3 = \frac{1 - 0.043 \times (0.961538 + 0.920936)}{1.043} = 0.881164$$

which are the discount factors of Example 7.6. The forward rates this curve implies are 4.0000%, 4.4088%, and 4.5136%.

Step 2: the forecast curve. The book also holds swaps referencing a twelve-month term index, quoted at 4.10% for one year and 4.32% for two. These are not OIS: the index is not the collateral rate, so the floating leg does not telescope and the forwards have to be solved for, while discounting on the Step 1 curve, which is already fixed. The one-year swap gives $F_1 = 4.1000\%$ directly. The two-year swap then gives F_2 from

$$0.0410 \times 0.961538 + F_2 \times 0.920936 = 0.0432 \times 1.882475$$

where $1.882475 = \mathrm{DF}_1 + \mathrm{DF}_2$ is the two-year annuity factor, so $F_2 = 4.5497\%$. The term index is forecast 10.00 basis points above the overnight rate in the first year and 14.09 in the second. The basis is not flat, and no single curve can carry both sets of forwards.

Step 3: the consistency check. A two-year basis swap pays the term index and receives the overnight rate plus a spread b. Setting its value to zero,

$$b = \frac{(0.0410 - 0.0400) \times 0.961538 + (0.045497 - 0.044088) \times 0.920936}{1.882475}$$

which comes to 0.001200, or 12.00 basis points. That is what the market quotes, so the pair of curves is consistent.

Notice what that check did *not* establish. With one schedule and one discount curve, b is forced to the difference between the two par rates, 4.32% − 4.20% = 12 basis points, and the basis swap has told the calibration nothing the outright quotes had not. Basis swaps are needed where that equality breaks (legs on different payment frequencies, or maturities and indices at which no outright swap trades), which is the usual reason a curve is pinned by basis quotes at the long end and by outright quotes at the short end.

Two practical warnings carry over from Chapter 4 and become sharper here. Interpolation choices matter more, because errors in one curve propagate into every other through the basis instruments. And the resulting object is not unique: two desks calibrating to identical quotes with different interpolation and different instrument selection will produce curves that agree at the quoted points and disagree between them. That disagreement is a genuine source of valuation difference, not a rounding artifact.

7.7 BASIS SWAPS

A **basis swap** exchanges two floating legs. Neither leg is fixed; the contract trades the spread between two indices, quoted as a number of basis points added to one leg to make the swap worth zero at inception.

Basis swaps serve two purposes. Commercially, they let an institution move exposure from one index to another without taking a directional rate position: a bank whose assets reference one index and whose liabilities reference another closes the mismatch with a basis swap, not with an ordinary swap. A euro borrower whose loan resets on three-month EURIBOR while its existing hedge pays six-month uses one to align the two, and a dealer that has written term SOFR

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swaps to loan borrowers lays off the term-against-overnight exposure the same way. Technically, they pin down the forecast curves of Steps 2 and 3 above wherever no outright swap on an index trades, and there the relationship between the curves would otherwise be unobservable. Within a single currency the euro 3s6s basis is now the main example; Section 33.9 returns to basis swaps as a product and to what their spreads price.

The spread is not arbitrary. It reflects the genuine economic difference between the two indices: differences in credit content, in the term over which the rate is fixed, and in the supply and demand for exposure to each. When the two indices are close substitutes the spread is small and stable; when they differ in credit character it widens under stress, exactly when it matters.

7.8 CROSS-CURRENCY SWAPS AND THE CROSS-CURRENCY BASIS

A **cross-currency swap** exchanges payments in two different currencies. Unlike a single-currency swap, the notional principals *are* exchanged, at the start and again at maturity, because the two legs are denominated in different money and there is nothing to net. Figure 7.5 shows the structure.



Figure 7.5: A EUR/USD cross-currency basis swap, from the perspective of a party raising dollars against euros. Notional is exchanged at both ends.

The economic content is a collateralized loan of one currency against another. A European bank needing dollars can borrow euros at home and swap them for dollars for five years, returning the dollars and reclaiming the euros at maturity. It is, in substance, secured dollar borrowing.

Three kinds of user account for most of the market. Banks outside the United States use it, as here, to fund dollar lending and dollar securities. Companies and supranational issuers borrow in whichever market is cheapest for them, often by issuing dollar or euro bonds, and swap the proceeds and the coupons back into the currency their revenues are in. And investors holding foreign bonds, such as a Japanese insurer holding US corporate debt, swap the foreign coupons and principal into their own currency so that the bond's return is not swamped by the exchange rate.

The cash flows are easiest to see in the fixed-for-fixed version, where neither leg floats, as in Example 7.9.

Example 7.9 Cash flows on a fixed-for-fixed currency swap

A European company raises dollars for five years. At a spot rate of 1.10 dollars per euro it pays €100 million and receives $110 million at the start, pays 4.2% on the dollars and receives 2.5% on the euros annually, and reverses the principal exchange at maturity. Amounts are in millions.

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Year	USD cash flow	EUR cash flow
0	+110.00	−100.00
1	−4.62	+2.50
2	−4.62	+2.50
3	−4.62	+2.50
4	−4.62	+2.50
5	−114.62	+102.50

The interest amounts are $110 \times 4.2\% = 4.62$ and $100 \times 2.5\% = 2.50$; year 5 adds the principal back.

Compare this with Example 7.1. Three differences stand out.

The principal is exchanged, twice. In a single-currency swap the two notionals are identical and cancel, so exchanging them would be pointless. Here they are in different money and cannot be netted, so they must actually change hands—and because they do, the contract carries exchange rate risk that an interest rate swap does not.

Nothing nets. Both legs are paid gross, in their own currency, on every date.

It is a bond issued and a bond bought. Read the dollar column alone and it is exactly the cash flow of issuing a $110 million 4.2% bond; the euro column is exactly that of buying a €100 million 2.5% bond. That is the whole instrument: borrow in one currency, lend in the other.

The exchange rate risk sits overwhelmingly in the final row. The annual interest amounts are small and partly offsetting, but the $114.62 million repayment against €102.50 million received is where a move in spot is felt, which is what Example 7.11 quantifies.

What actually trades in volume is the floating-for-floating version of Figure 7.5, where each leg pays its own currency's overnight rate and the whole price of the transaction is compressed into a single spread: the basis.

Two distinctions matter in practice. Between dealers the standard contract is the **mark-to-market** (resettable) cross-currency swap, in which the dollar notional is reset to the current spot equivalent at each payment date and the difference is paid in cash, so the exchange rate exposure that Example 7.11 measures is settled period by period instead of accumulating to maturity; Section 33.9 prices the difference. The constant-notional swap of the examples here is what a company hedging a fixed-size bond issue or foreign investment uses. And a cross-currency swap is not an **FX swap**, which is a spot exchange of currencies reversed by a forward exchange, with no interest payments in between: the interest differential is built into the forward points of Section 5.7. FX swaps are by far the larger market, concentrated at maturities under a year, and are how banks and investors fund and hedge foreign-currency positions from day to day; cross-currency swaps carry the same funding trade out to many years.

The cross-currency basis

Textbook theory says the two floating legs should exchange flat: covered interest parity holds, and any spread would be arbitrageable by borrowing in one currency, swapping, and lending in the other.

Since 2008 this has been persistently false. A spread, the **cross-currency basis**, is quoted on one leg, and for the euro, yen, Swiss franc and sterling against the dollar it has been negative for most of the period since, meaning the party *providing dollars* is paid a premium. (The Australian and New Zealand dollars, whose banks and companies borrow abroad and swap the proceeds home, have often run a positive basis: there the scarce side is the local currency.) The basis widened dramatically during the 2008 crisis and again during the 2011–12 European sovereign crisis, and although it narrowed afterward it did not return to zero. It also spikes predictably at quarter-ends and year-ends.

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Case Study 7.2 The end of covered interest parity

Covered interest parity is not a theory about expectations or risk preferences. It is an arbitrage relationship: borrow dollars, convert to euros at spot, invest in euros, and sell the euro proceeds forward for dollars. If the resulting dollar return differs from the dollar interest rate, someone has been handed a riskless profit.

For decades the relationship held to within a basis point or two. Since 2008 it has failed continuously—not by a basis point but by tens of basis points, for years at a time, in the most liquid currency pairs in the world.

The explanation is not that traders stopped noticing, and it has two phases. In 2008 and again in 2011–12 the basis measured stress: banks outside the United States needed dollars, their credit was in doubt, and dollar lenders would supply them only at a premium. Dollar swap lines between the Federal Reserve and other central banks, opened in 2008 and widened again in March 2020, are what capped those spikes. The persistence afterward has a different cause: the arbitrage requires balance sheet. Post-crisis regulation (the leverage ratio of Section 1.12 in particular, phased in from 2015) made it expensive for banks to expand their balance sheets to capture small spreads. Against that limited supply sits steady demand: Japanese and European banks, insurers and pension funds holding dollar assets fund or hedge them through FX and cross-currency swaps, and they pay the premium. The trade is still there; no one wants to hold it. This is also why the basis widens at reporting dates, when balance sheet is scarcest.

The episode is an instance of the limits of arbitrage set out in Section 5.15.

Example 7.10 The cost of a negative basis

A European company swaps €100 million into dollars for five years at a cross-currency basis of −20 basis points. It has lent euros and borrowed dollars, so it pays SOFR flat on the dollar leg and *receives* €STR less 20 basis points on the euro leg. Receiving 20 basis points less than the euro overnight rate is precisely how it pays for the dollars.

With euro discount factors of 0.975610, 0.951814, 0.928599, 0.905951, and 0.883854, the euro annuity is

$$A_{\text{EUR}} = 4.645828$$

The present value of the 20 basis point spread is

$$100{,}000{,}000 \times 0.0020 \times 4.645828 = €929{,}166$$

So the negative basis is worth just under a million euros on this trade, paid by the party that wanted dollars, earned by the party that supplied them. It is precisely the quantity that a single-curve valuation would set to zero.

Valuation

A cross-currency basis swap is most easily valued as a pair of floating-rate notes: long a note in the currency lent, short a note in the currency borrowed, with the two converted at the current spot rate. At a reset date each note prices at par *provided* its spread equals the current market basis; when the market basis has moved, the off-market spread is worth its difference times the annuity, exactly as in Section 7.3.

Example 7.11 Valuing a cross-currency swap after the market moves

The company of Example 7.10 entered its five-year swap three years ago: it lent €100 million and borrowed $110 million at a spot rate of 1.10 dollars per euro, receiving €STR less 20 basis points.

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Two years remain. The euro discount factors for those two years are 0.975610 and 0.951814, so the remaining annuity is $A_{\text{EUR}} = 1.927424$.

Case 1: only the exchange rate has moved. Spot is now 1.05, and the market basis is still −20 basis points. Because the contract's spread equals the market spread, the euro note prices at par and so does the dollar note:

$$V = €100{,}000{,}000 - \frac{\$110{,}000{,}000}{1.05} = 100{,}000{,}000 - 104{,}761{,}905 = -€4{,}761{,}905$$

The company borrowed dollars and the dollar has strengthened, so the liability has grown in euro terms.

Case 2: the basis has also widened. Suppose in addition the market basis has moved to −45 basis points. The contract pays €STR less 20, where the market now requires €STR less 45, so the euro note pays 25 basis points *more* than the market demands and is worth more than par. The company receives that leg, so the move is in its favor:

$$100{,}000{,}000\,(1 + 0.0025 \times 1.927424) = €100{,}481{,}856$$

and the swap is worth

$$100{,}481{,}856 - 104{,}761{,}905 = -€4{,}280{,}049$$

The widening basis has offset €481,856 of the exchange rate loss, because the company locked in dollar funding when dollars were cheaper than they are now.

Two lessons follow. The exchange rate move is an order of magnitude larger than the basis move (€4.76 million against €0.48 million), which is why cross-currency swap books are managed first as FX exposures and only then as basis exposures. And the basis is a genuine mark-to-market risk in its own right: a position that is fully hedged in both currencies' rates and in spot FX still moves when the basis does.

Valuation also requires a decision about the collateral currency, because Section 7.4 applies here too. A trade collateralized in dollars and a trade collateralized in euros have different values even with identical cash flows. Many CSAs permit the poster to choose among several eligible currencies, which makes the collateral a rolling option to deliver whichever currency is cheapest, an option that must itself be valued, and one that Section 8.5 takes up where it belongs, alongside the rest of collateral practice.

7.9 SWAP SPREADS AND THE PRICE OF BALANCE SHEET

The **swap spread** is the difference between the swap rate and the yield on a government bond of the same maturity. For most of the swap market's history it was positive, and the explanation was obvious: the swap curve carried bank credit risk through its LIBOR index, while the government curve did not, so the swap rate stood above the government yield by roughly the market's assessment of that credit difference.

Dollar swap spreads have now been negative for years: at thirty years since late 2008, at ten years since the middle of the 2010s, and, since the switch to SOFR, across most of the curve. The switch itself lowered the series, because a SOFR swap carries no bank credit at all, so spreads measured before and after 2021 are not directly comparable. Read literally, this says the market accepts a lower fixed rate from a bank than from the sovereign (that banks are better credits than the government), which would be an extraordinary claim, and is not what it means.

The resolution is the argument of the previous section, transplanted. Example 7.12 sets out the trade a negative swap spread appears to offer.

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Example 7.12 The arbitrage no one wants

The ten-year swap rate is 4.00% and the ten-year government bond yields 4.10%, so the swap spread is −10 basis points. The apparent trade is:

1. Buy $100 million of the ten-year bond, yielding 4.10%, funding it in the repo market at approximately the overnight rate.
2. Pay fixed at 4.00% in a ten-year swap, receiving the compounded overnight rate.

The floating leg received on the swap approximately offsets the repo cost of funding the bond. What remains is a fixed 4.10% received against a fixed 4.00% paid: 10 basis points, apparently for no market risk at all. With a ten-year annuity of 8.111 at 4%, the present value is

$$100{,}000{,}000 \times 0.0010 \times 8.111 = \$811{,}100$$

The trade is real, and it sat available for years. What it requires is a balance sheet: the bond must be held and financed. Under the leverage ratio (Section 1.12), holding $100 million of government bonds consumes capital whether the position is hedged or not. A dealer weighing $811,100 against the capital that position ties up for a decade may reasonably decline.

So a negative swap spread is not a statement about relative creditworthiness. It is the price of the balance sheet needed to close it—the same phenomenon, measured a different way, as the cross-currency basis of Section 7.8. Heavy government bond supply pushes yields up; regulation makes the offsetting arbitrage expensive; and demand from pension funds and insurers to receive fixed in swaps, which requires almost no balance sheet, pushes swap rates down. The spread settles wherever those forces balance, and there is no reason that point should be positive.

Two recent regulatory changes bear directly on it. In November 2025 the US banking agencies recalibrated the enhanced supplementary leverage ratio for the largest banks, in force since April 1, 2026, with the stated aim of making it a backstop rather than a binding constraint; the more balance sheet that releases for holding Treasuries, the less negative swap spreads need be. And the SEC's Treasury clearing mandate of Section 2.11, which applies to cash trades from December 31, 2026 and to repo from June 30, 2027, lets dealers net cleared Treasury and repo positions against one another, which also reduces the balance sheet the trade consumes. Relative-value funds that do put the trade on, financed in repo, carry the two risks a dealer avoids: that the spread moves further against them before it converges, and that their repo funding is withdrawn. In March 2020 both happened at once.

Swap spreads, like the cross-currency basis, are the limits of arbitrage of Section 5.15 at work, and they can stay dislocated for years without anyone behaving irrationally.

7.10 CREDIT RISK, NETTING, AND CLEARING

A swap creates credit exposure, but of an unusual shape. Unlike a loan, where one party is always the creditor, a swap can have positive value to either side depending on where rates go. Exposure is therefore *two-sided* and changes sign over the contract's life.

It is also far smaller than the notional suggests. The notional is never exchanged; what is at risk is the swap's replacement cost (what it would cost to replace the contract on current terms if the counterparty defaulted), which by (7.6) is $L(S - K)A$. A $100 million ten-year swap that has moved 50 basis points in your favor exposes you to roughly $4 million, not $100 million. This is why aggregate notional statistics (hundreds of trillions) are a poor guide to systemic exposure.

Three mechanisms reduce what remains.

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Netting. The ISDA Master Agreement provides for close-out netting, which Section 1.5 and Example 1.1 set out, cherry-picking included. Netting enforceability is a legal question decided jurisdiction by jurisdiction, which is why China's Futures and Derivatives Law, in force since August 2022, mattered so much to the dealers trading with Chinese counterparties: it was the first Chinese statute to recognize close-out netting. Until a jurisdiction settles the question, the netted figure a dealer reports is not necessarily the figure that would survive its counterparty's bankruptcy court.

Collateral. A credit support annex requires the out-of-the-money party to post collateral against current exposure, reducing it toward zero between margin calls. Section 7.4 showed the valuation consequence; Chapter 8 examines what posting collateral does to liquidity, which is a different and sometimes larger problem.

Central clearing. Post-crisis reforms require standardized swaps to be cleared, which novates each one to a central counterparty by the mechanism of Section 2.4. Bilateral credit risk is replaced by exposure to the CCP, and Chapter 2 examines what that concentration achieves and what it costs.

The same reforms changed how swaps are traded and recorded. In the United States, standard swaps that the regulator has declared "made available to trade" must be executed on a swap execution facility (SEF), an electronic venue where requests for quotes go to several dealers at once, and the European Union imposes a matching trading obligation. Every swap, cleared or not, is reported to a trade repository, which is what makes the market's size measurable at all. Dealers periodically take part in portfolio compression, tearing up offsetting trades among many participants to cut gross notional without changing anyone's net risk (Section 33.12). And swaps that are not cleared carry mandatory initial margin under the uncleared margin rules of Section 8.4, which is deliberately more expensive than clearing and is one of the main reasons standard trades clear.

What survives all three is the residual counterparty risk priced as a credit valuation adjustment (CVA) in Chapter 9 and capitalized under the framework of Chapter 27. It is smaller than it once was, but it is not zero, and the 2021 episode discussed in Chapter 27 shows how quickly it returns when a portfolio is concentrated and margining is lax.

7.11 WHAT THE TRANSITION DID TO THE SWAP BOOK

Chapter 4 described the ISDA fallback mechanism in the abstract. It is worth seeing what it meant for an actual portfolio, because the swap book was where the transition would succeed or fail.

A dealer entering 2021 held millions of swaps referencing LIBOR, with maturities running decades out. Three things had to happen. New trades had to move to RFR references, which market convention accomplished by coordinated "big bang" dates on which interdealer trading switched. Cleared legacy trades were converted by the clearing houses before each LIBOR setting ended (sterling, yen and Swiss franc in December 2021, dollars in April and May 2023): each LIBOR swap was replaced by a standard OIS, with the fallback spread adjustment folded into the fixed rate and cash paid for the remaining differences in convention. Bilateral legacy trades fell back under the protocol.

The two routes left different things behind. A converted cleared swap is an ordinary OIS, indistinguishable from one traded new. A bilateral LIBOR swap that fell back references compounded SOFR plus the fixed spread of Table 4.2 for the rest of its life, under the protocol's own conventions (Example 4.1), so those spreads remain a structural feature of the legacy book. Because the spread is a constant over compounded SOFR, such a trade needs no curve of its

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own; what it needs is a booking and confirmation system that records it correctly, and the stock of these trades shrinks only as they mature or are restructured.

The broader observation is the one from Chapter 4, now visible at portfolio scale: the transition did not eliminate the multi-curve structure. It changed which curves the structure contains.

7.12 OTHER SWAPS

The vanilla structure generalizes in many directions, of which a few deserve naming here; Chapter 33 treats them in full.

Amortizing and accreting swaps vary the notional over time, typically to match an underlying loan (Section 33.1). **Forward-starting swaps** begin at a future date (Section 33.2); a company planning to issue a bond in six months pays fixed on one today to lock in the rate, and they are the underlying for the swaptions of Chapter 29. **Constant maturity swaps** reference a long-tenor swap rate on each reset; Section 33.5 describes them and Section 30.5 derives the convexity adjustment they need. **Inflation swaps** exchange a fixed rate for the realized change in a consumer price index and are how pension funds hedge inflation-linked liabilities (Section 33.10).

Equity swaps exchange the total return on an equity or index for a floating rate. **Total return swaps** extend the same exchange to any reference asset (a bond, a loan portfolio, a single stock), and they transfer the full economics of an asset without transferring the asset. Hedge funds use them to hold exposure they would otherwise have to finance and disclose, and banks provide them as a form of secured lending, which is why they are the instrument through which leverage accumulates quietly. Section 33.11 treats both, and Chapter 27 examines what total return swaps made possible in 2021.

Commodity swaps exchange a fixed price for a floating commodity index (Section 33.8), and are how airlines fix their fuel cost and producers their selling price for years ahead; **variance swaps** exchange a fixed variance for realized variance and are treated in Chapter 22. **Credit default swaps** (CDS), despite the name, are structurally quite different (one leg is a periodic premium, the other pays only if a specified issuer defaults) and belong to Chapter 25.

What unites them is the mechanism of this chapter: identify the cash flows, project the floating ones on the appropriate forecast curve, and discount all of them on the curve the collateral agreement dictates.

SUMMARY

An interest rate swap exchanges fixed for floating payments on a notional that is never exchanged; in most major currencies the standard contract is now an overnight indexed swap referencing a compounded risk-free rate in arrears, while euro swaps still reference EURIBOR. Valued as a portfolio of forward rate agreements, it is worth $L(S - K)A$, the gap between the current par rate and its own fixed rate, annuitized. The comparative advantage argument traditionally used to motivate swaps does not survive inspection, because the fixed and floating borrowing rates it compares are not comparable commitments: the floating lender retains the right to reprice or withdraw, and the apparent gain is compensation for that. Swaps are better explained by their ability to separate the interest rate character of an exposure from the funding decision that created it.

The discount curve is not a modeling choice. For a collateralized derivative the correct discount rate is the rate paid on its collateral, because that is what replicating the position costs; the same cash flows are therefore worth different amounts under different credit support annexes, which is why the move to SOFR discounting in October 2020 required coordinated compensation across cleared markets. Valuation needs at least two curves: a discount curve

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determined by the CSA and one or more forecast curves determined by the indices the contracts reference. Where the two coincide the floating leg telescopes to $L(1 - \mathrm{DF}_n)$ and the familiar single-curve results follow; where they do not, that identity fails and the error equals the basis times the annuity. Curves are built discount first, and in practice by simultaneous calibration.

Basis swaps exchange two floating legs, hedge the mismatch between indices, and are the instruments that pin down the forecast curves. Cross-currency swaps exchange payments and notionals in two currencies and are economically secured borrowing across currencies; their basis has been persistently nonzero since 2008, first as a measure of dollar funding stress and then because the arbitrage that would close it consumes balance sheet that post-crisis regulation made expensive, which is also why it widens predictably at reporting dates. Negative long-dated dollar swap spreads are the same phenomenon measured a second way: closing that gap means financing a bond, and the leverage ratio charges capital for it regardless of the position's risk.

Swaps create two-sided credit exposure far smaller than the notional, being the replacement cost $L(S - K)A$ rather than the principal. Close-out netting under the ISDA Master Agreement collapses a defaulted counterparty's trades to a single net amount and prevents cherry-picking; collateral under a CSA reduces the residual toward zero between margin calls; and central clearing replaces bilateral exposure with exposure to a CCP that nets multilaterally. What remains is priced as CVA (Chapter 9) and capitalized under the framework of Chapter 27.

FURTHER READING

Bianchetti's "Two Curves, One Price" (*Risk*, August 2010) is an early statement of the multi-curve framework, and Piterbarg's "Funding Beyond Discounting" (*Risk*, February 2010) derives the collateral-rate discounting result that the framework rests on. Henrard's *Interest Rate Modelling in the Multi-Curve Framework* is the book-length treatment. Du, Tepper, and Verdelhan (2018) is the standard reference on the cross-currency basis, and it is the empirical counterpart to this chapter's claim that the basis is a price rather than an error.

PRACTICE QUESTIONS

7.1 State the comparative advantage argument for swaps, explain the flaw in it, and give the better explanation for the size of the swap market.

7.2 A two-year annual swap on a notional of \$50 million pays fixed at 3.50% and receives floating. Discount factors are 0.9709 and 0.9426; forward rates are 3.00% and 3.30%. Value the swap to the fixed payer and find the par swap rate.

7.3 Show that when the floating index and the discount curve are the same, the floating leg is worth $L(1 - \mathrm{DF}_n)$, and state precisely the condition under which this fails.

7.4 Explain why the correct discount rate for a fully collateralized derivative is the rate paid on its collateral, and state what follows about two otherwise identical swaps collateralized in different currencies.

7.5 Repeat Question 7.2 assuming the floating index has forward rates 15 basis points above those given, with discount factors unchanged, and verify that the change in value equals the basis times the annuity.

7.6 Describe the order in which curves must be built in a multi-curve framework, explain why the discount curve must come first, and explain why a single currency needs more than one forecast curve.

7.7 Explain what a basis swap is, give two distinct reasons the market needs them, and explain why notional principals are exchanged in a cross-currency swap but not in a single-currency one.

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7.8 The eight-year swap rate is 3.60% and the eight-year government bond yields 3.72%. Compute the swap spread, describe the trade it appears to offer, calculate its present value on a $250 million position using an eight-year annuity of 6.85, and explain both what prevents the trade from being done and why a negative spread does *not* imply that banks are better credits than the government.

7.9 Explain why a swap creates two-sided credit exposure and why the exposure is far smaller than the notional. For the swap of Question 7.2, state the exposure of each party.

7.10 Explain what close-out netting achieves, what would happen without it, and why its legal enforceability is a jurisdiction-by-jurisdiction question.

FURTHER QUESTIONS

7.11 A five-year cross-currency swap on a notional of €200 million carries a basis of −35 basis points. Using the euro discount factors of Example 7.10, find the present value of the basis. Who pays it, and what are they paying for?

7.12 Covered interest parity is an arbitrage relationship, not a behavioral theory. Explain carefully why it has failed persistently since 2008, and why the failure is largest at quarter-ends and year-ends.

7.13 When clearing houses switched dollar discounting from the federal funds rate to SOFR in October 2020, they paid cash compensation to members and also distributed basis swaps. Explain what each of the two was compensating for, and why cash alone would not have been sufficient.

7.14 A dealer's swap book is flat to a parallel shift in rates but holds large positions in different tenor buckets and in two currencies. Identify at least three distinct risks the book still carries that Chapter 4's single-curve framework could not have represented.

7.15 A CSA permits the collateral poster to deliver any of three currencies. Explain why this makes the collateral an option, why that option has value, and how you would begin to quantify it. Which chapter's machinery would you need?

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